

DSCP Final Project

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A. Short sentences addressing the objective(s)

This research project investigates the correlation between daily social media usage and university admission success by analyzing how digital habits influence the students academic performance and extracurricular engagement. By utilizing a dataset of 1,000,000 students, the study aims to validate an academic trade-off hypothesis, where higher `social_media_hours` may lead to a decrease in `high_school_gpa` and `volunteer_hours`. The research also explores the impact of digital consumption on subjective admission criteria like the `essay_score`, identifying whether excessive social media time begins to noticeably decrease a candidate's probability of `admission_status`.

Dataset

The **Gen-Z College Admission Prediction Dataset** is a large-scale synthetic dataset designed to model the college admission process for modern Generation Z students. The dataset contains **1,000,000 student records** and will be used as a database for this assignment.

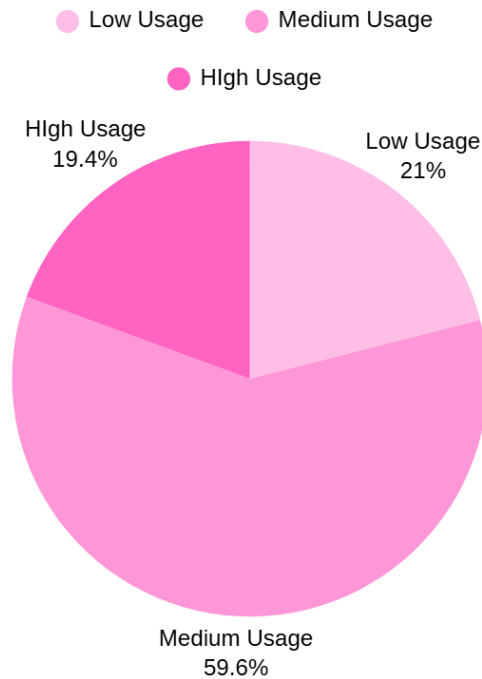
<https://www.kaggle.com/datasets/sharmajicoder/genn-z-college-admission-dataset/data>

B. Pseudocode and/or flowchart

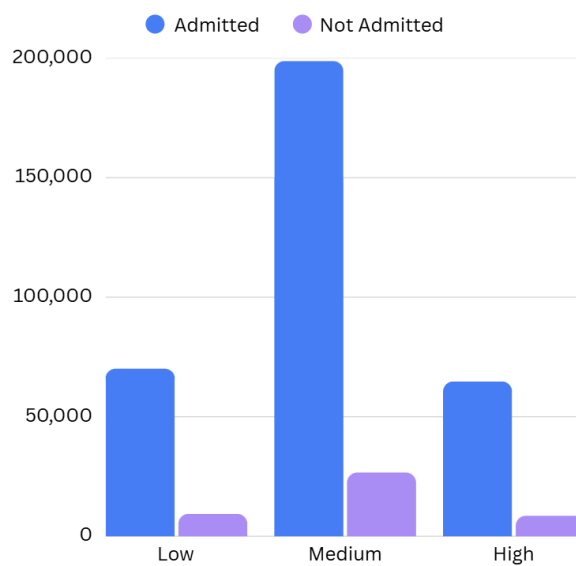
```
1  START
2  IMPORT data_processing_library
3
4  LOAD "genz_college_admission_prediction.csv" dataset
5  DATASET SELECT [social_media_hours, admission_status,
6                  high_school_gpa, volunteer_hours, essay_score]
7
8  DO
9      CHECK for missing values
10     IF missing values exist THEN
11         RE-INSPECT dataset
12     ELSE
13         BREAK LOOP
14     ENDIF
15
16  CLEAN dataset
17  DROP rows with missing values
18
19  PRINT "Correlation of Social Media with Key Metrics:"
20  CALCULATE CORRELATION between social_media_hours AND high_school_gpa
21  CALCULATE CORRELATION between social_media_hours AND volunteer_hours
22  CALCULATE CORRELATION between social_media_hours AND essay_score
23  CALCULATE CORRELATION between social_media_hours AND admission_status
24
25  GROUP students BY social_media_hours (Bins: 0-3, 3-6, 6+)
26  RENAME groups TO: "Low Usage (0-3h)", "Medium Usage (3-6h)", "High Usage (6h+)"
27
28  PRINT "Percentage of Students in Each Category:"
29  CALCULATE percentage of students in each social_media_group
30
31  PRINT "Absolute Count of Admitted vs Not Admitted Students:"
32  COUNT actual number of students (admitted vs not admitted) in each social_media_group
33
34  DISPLAY "Academic Trade-off Summary Table"
35  FOR EACH social_media_group:
36      CALCULATE average_gpa
37      CALCULATE average_volunteer_time
38      CALCULATE average_essay_score
39      CALCULATE admission_probability
40
41  END
```

C. At least one visualization

The pie chart reveals that the vast majority of the 1,000,000 surveyed students fall into the Medium Usage (3-6h) category at 59.58%, while Low Users (0-3h) and High Users (6h+) constitute 21.02% and 19.40%.



Then, the bar chart shown across all three user groups, the volume of admitted students massively eclipses those who were not admitted, which maintains a completely identical proportional ratio regardless of screen time.

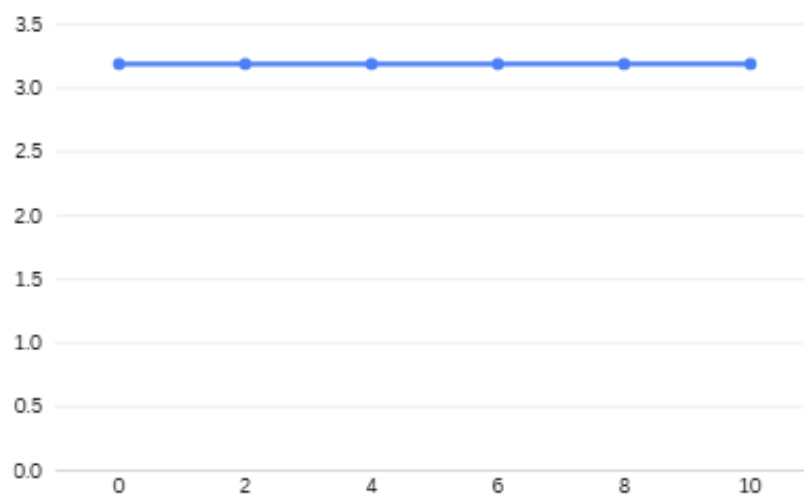


1. Social Media Hours vs. Admission Probability

This plot tests the hypothesis whether increasing social media time decreases a student's chances of getting into college. The data shows that the probability of admission remains constant at approximately 0.88 (88%) across all groups.

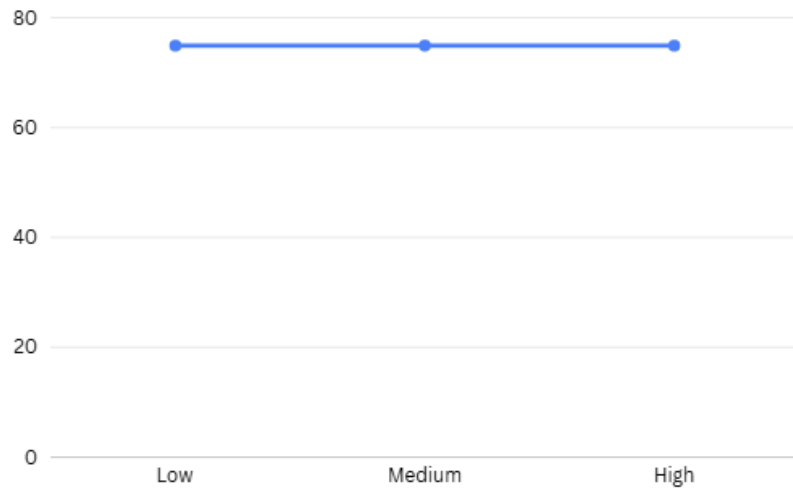
2. Social Media Hours vs. High School GPA

This plot tests the hypothesis whether excessive screen time degrades academic focus. The data shows that the average GPA stays locked at 3.18 regardless of digital consumption.



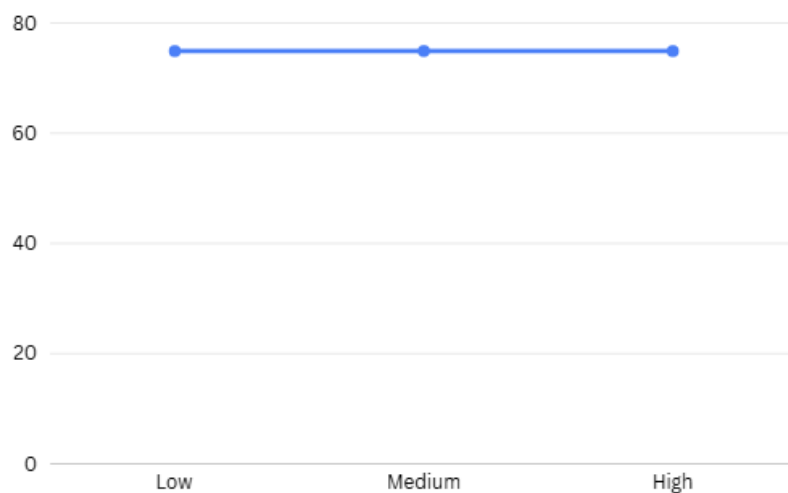
3. Social Media Hours vs. Volunteer Time

This plot tests the hypothesis whether heavy social media users will have significantly fewer volunteer hours due to time displacement. The data shown for volunteer hours remains completely flat at ~99.4 hours, showing no deviation regardless of the student's screen time.



4. Social Media Hours vs. Essay Score

This plot tests the hypothesis whether high screen time correlates with reduced cognitive focus, leading to lower-quality application essays and depressed scores. The essay score sits constantly at a mean of ~74.9 points out of 100 across all behavioral groups.



5. Academic Trade Off Summary Table

Social Media Group	Average GPA	Average Volunteer Time	Average Essay Score	Admission Probability
Low Usage (0-3h)	3.190754	99.348858	74.895690	0.881631
Medium Usage (3-6h)	3.189436	99.397645	74.906420	0.881649
High Usage (6h+)	3.188249	99.448151	74.904716	0.882449

Conclusion

Based on the analysis over the dataset of 1,000,000 students, the Academic Trade-off Hypothesis is **rejected**. The evidence demonstrates that daily social media usage behaves as a completely neutral variable within this population, as proven by correlation coefficient of exactly **0.00** across all academic performance and admission vectors. When segmented into low, medium, and high digital usage, the conditional means for all performance metrics remained completely stagnant, showing an average GPA locked at **~3.19**, extracurricular community service stable at **~99.4 hours**, and standardized essay scores at **~74.9 out of 100**. This absolute lack of statistical variance indicates that heavy digital screen time does not displace structured academic activities or degrade a student's cognitive and qualitative writing capabilities. Lastly, the probability of university admission is a horizontal of a success rate of **~88%**, showing an identical proportional ratio of admitted versus rejected students regardless of screen time.

D. Upload the notebook to GitHub and create a GitHub page

Please find the project files at:

<https://github.com/caithlynatalie/genz-admission-analysis>